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SPECIFIC
TECHNICAL

Gas measurement systems

Hourly gas measuring devices

Part 1: General characteristics of the remote management or remote reading system

UNI/TS 11291-1

May 2013

Gas measurement systems

Hourly based gas metering systems

Part 1: General requirements for remote reading or remote management system

The technical specification defines the architecture of the remote reading or remote management system for the measurement groups of gas transport and distribution networks, introducing requirements relating to the minimum functions that must be performed by the devices that comprise it and which allow the measurement of gas volumes, the conversion of gas volumes to reference thermodynamic conditions, the recording of measurement data also useful for balancing the transport and distribution networks and their remote transmission.

ITALIAN TEXT

This technical specification replaces UNI/TS 11291-1:2010.

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PREMISE

Compared to the previous edition, the following changes have been made:

1) updated regulatory references;

2) the necessary alignment with the UNI EN 12405-1 standard was carried out;

3) updated point 4.3.1 for aspects relating to device synchronization;

4) new points for the management and maintenance of the equipment have been inserted;

5) the bibliography has been integrated.

The technical specification is reviewed every three years. Any comments on UNI/TS 11291-1 must be received by UNI by May 2015.

This technical specification was developed under the jurisdiction of the body federated to UNI

CIG - Italian Gas Committee

The UNI Central Technical Commission gave its approval on 23 April 2013.

This technical specification was ratified by the President of UNI and became part of the national regulatory body on 23 May 2013.

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INTRODUCTION

This technical specification introduces requirements relating to the remote reading or remote management system and the devices that comprise it and which allow the measurement of gas volumes, the conversion of gas volumes to reference thermodynamic conditions, the recording of measurement data also useful for balancing distribution networks and their remote transmission.

On October 22, 2008, the Italian Electricity and Gas Authority published Resolution ARG/gas 155/08, "Guidelines for the commissioning of gas metering units, characterized by minimum functional requirements and with remote reading or remote management functions, for delivery points of natural gas distribution networks." Amendments to the text of this technical specification may be necessary as a result of measures adopted by the Authority following this resolution.

With regard to the volume conversion function, this specification refers to UNI EN 12405-1 "Gas meters - Conversion devices - Part 1: Volume conversion". The same standard is also used to indicate some requirements that gas meters must meet so that the periodic recordings derived from them comply with this specification.

As regards data transmission, the UNI CEI EN 13757 "Communication systems for meters and remote reading of meters" and the CEI EN 60870 series "Remote control systems and equipment" were considered.

The system's functionalities were taken into consideration in accordance with NTA 8130 - Dutch standard for smart metering.

1 PURPOSE AND SCOPE

This technical specification establishes:

- what is the system architecture that can be used for remote reading or remote management of metering units;
- the services and functions that must be made available by the system;
- the minimum performance that the system must guarantee to ensure the services and functions;
- the minimum performance that each of the devices considered must guarantee and the minimum construction requirements it must meet to guarantee said performance, referring the details of the functional and construction requirements to the other parts of the UNI/TS 11291 series;
- the methods for remote data transmission;
- the installation and maintenance operations of the various devices necessary to guarantee the specified performance over time.

This technical specification applies to systems for the remote reading or remote management of GdMs serving the Delivery Points of gas distribution networks.

The aim is to make all devices (excluding the centers) that operate in the remote reading or remote management system interoperable.

2 REGULATORY REFERENCES

This technical specification refers, through dated and undated references, to provisions contained in other publications. These normative references are cited at the appropriate points in the text and are listed below. For dated references, subsequent amendments or revisions made to those publications are valid only if introduced into this technical specification as an update or revision. For undated references, the latest edition of the referenced publication (including updates) applies.

UNI/TS 11291-2	Gas measurement systems - Hourly gas measurement devices - Part 2: CTE protocol
UNI/TS 11291-3	Gas measurement systems - Hourly gas measurement devices - Part 3: CTR protocol

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UNI/TS 11291-4	Gas measurement systems - Hourly gas measurement devices - Part 4: Requirements for flow rate measurement units > 65m3/h (counter >G40)
UNI/TS 11291-5	Gas measurement systems - Hourly gas measurement devices - Part 5: Requirements for 16 m range measuring units3/h up to 65 m3/h (counter≥G10 and≤G40)
UNI/TS 11291- 6	Gas measurement systems - Hourly gas measurement devices - Part 6: Requirements for measuring sets with ranges less than 10 m3/h (counter <G10)
UNI/TS 11291-7	Gas metering systems - Hourly gas metering devices - Part 7: Remote management systems for gas meters - SAC, Concentrators, Repeaters and Translators
UNI/TS 11291-8	Gas measurement systems - Hourly gas measurement devices - Part 8: Protocols for remote management of metering units for the distribution network
UNI/TS 11291-10	Gas measurement systems - Hourly gas measurement devices - Part 10: Safety
UNI EN 13757-1	Communication systems for meters and remote meter reading – Part 1: Data exchange
UNI EN 12405-1:2007	Gas meters - Conversion devices - Part 1: Volume Conversion
UNI EN ISO 9001	Quality Management Systems - Requirements

3 TERMS, DEFINITIONS, SYMBOLS AND ABBREVIATIONS

3.1 Terms and definitions

For the purposes of this technical specification, the terms and definitions in UNI EN 12405-1:2007 and the following terms and definitions apply.

3.1.1 **gas year**Conventional year of 12 months duration from which the conventional periods begin which starts from the first gas day of the conventional month (for example October).

3.1.2 **apparatus**: Physical component of the measurement system used to convert and/or record and/or communicate data. The device can perform one or more functions.

3.1.3 **remote reading or remote management center**: Set of central hardware and software components intended for managing the remote reading and remote management functions of the measurement groups, as well as for managing the communication functions.

3.1.4 **GdM class**: Class, identified by letters A1, A2, A, B and C depending on the field of application, the functions and the measurement data that the GdM itself must detect, record locally and possibly transmit remotely.

3.1.5 **end customer**: Consumer who purchases gas for his own use.

3.1.6 **data concentrator**: Intelligent peripheral device intended for collecting sampling data, processing it and retransmitting it to the SAC; it processes and, if necessary, forwards the commands received from the SAC to the GdM.

3.1.7 **reference thermodynamic conditions**: Conditions used to express the volume of gas independently of the actual pressure and temperature conditions at the point of measurement.

3.1.8 **volume conversion**: Function of the GdM capable of bringing the measured gas volumes back to the reference thermodynamic conditions.

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3.1.9 **device**: Generic device enabled to operate in the remote reading or remote management system.

3.1.10 **measurement data**: Data that the GdM must collect, record locally and transmit remotely.

3.1.11 **tariff band**: Time interval of a calendar day delimited by a (whole) start hour and a (whole) end hour in which a single rate (or the same price) is in force.

3.1.12 **gas day**: Conventional day of 24 hours duration that starts from a conventional full hour (for example 06:00).

3.1.13 **measurement group (GdM)**For the distribution network, it is the part of the end customer's supply system used for gas interception, measurement, and connection to the customer's internal system. The measurement group consists of one or more units, which, regardless of the technology used, are capable of measuring, converting measured volumes, communicating, and managing the gas supply.

3.1.14 **significant event**: Event or report in the GdM, which can affect its functioning or alter the data contained therein.

3.1.15 **HAN (Home Area Network)**: Data communication network that connects digital devices in home environments.

3.1.16 **interoperability**: Ability to operate products of different types and from different manufacturers in the same network under the same rules and conditions.

3.1.17 **MAN (Metropolitan Area Network)**: A computer network that usually covers an entire city.

3.1.18 **measurement**: GdM function that allows you to detect gas consumption.

3.1.19 **multi-access**: Property of a GdM that enables access to the same GdM by multiple remote management centers for configuration and reading, in this case even simultaneously. The access policy and related authorizations are managed through a set of configurable rules.

3.1.20 **NAN**: Data communication network that connects devices installed in nearby buildings in an urban setting.

3.1.21 **local time**: Time calculated from UTC taking into account the time zone and daylight saving time when in effect.

3.1.22 **billing period**: A period of time that always begins on the first gas day of a month and ends on the first gas day of one or more subsequent months; it can be 1, 2, 3, 4, or 6 months.

3.1.23 **tariff period**: Period of time in which the same rate applies.

3.1.24 **conventional flow rate**: Flow rate calculated every 5 min as the average of the volumes at the reference thermodynamic conditions computed in the last 15 min.

3.1.25 **deliverable flow rate**: Maximum gas flow rate, at the reference thermodynamic conditions, for which the GdM is sized.

3.1.26 **tariff program**: Scheme of time intervals, for each calendar day, to which the different unit prices of the tariff apply. It is used to allocate consumption in real time in order to comply with the tariff requirements.

3.1.27 **data logging**: GdM function that allows you to acquire measurement data, process them and record them at a pre-set frequency.

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3.1.28 **repeater**: System device that extends the physical length of the network without changing the transmission protocol.

3.1.29 **SAC (Central Acquisition System)**: Functional module of the remote reading or remote management system that handles network management and access to field devices.

3.1.30 **remote reading or remote management system**: Set of the remote reading or remote management center, the data concentrators, the measurement groups and the devices necessary to ensure communication.

3.1.31 **application software**: All software parts that have no metrological impact.

3.1.32 **rate**: Set that establishes the elements considered and the calculation methods relating to the amounts owed by the user, based on the characteristics of the supply relating to each user or category of users.

3.1.33 **multi-hourly rate**: Tariff that provides differentiated prices according to daily time periods, daily periods of the week and periods of the year.

3.1.34 **remote management**: Set of IT tools and functions that allow the remote start-up, preparation, and management of measurement groups installed at delivery points of gas distribution networks.

3.1.35 **remote reading**: Remote management which does not include the possibility of sending commands for supply management.

3.1.36 **translator**: System device that performs a transformation of the communication protocol.

3.1.37 **transmission**: GdM function that allows remote transactions to be made with the remote reading or remote management center.

3.1.38 **UTC**Coordinated Universal Time, which is the reference time zone from which all other time zones in the world are calculated.

3.2 Symbols and abbreviations

For the purposes of this technical specification the following abbreviations apply.

3.2.1 **AES128**-Advanced Encryption Standard for Computer Security Standard (NIST FIPS PUB 197) with 128-bit key.

4 REMOTE READING OR REMOTE MANAGEMENT SYSTEM FOR GAS METERS

4.1 System architecture

The architecture of the system for the remote reading and management of measurement groups intended for the delivery points of gas distribution networks is illustrated in figure 1 and may include Measurement Groups (GdM), Repeaters, Protocol Translators, Concentrators and centers.

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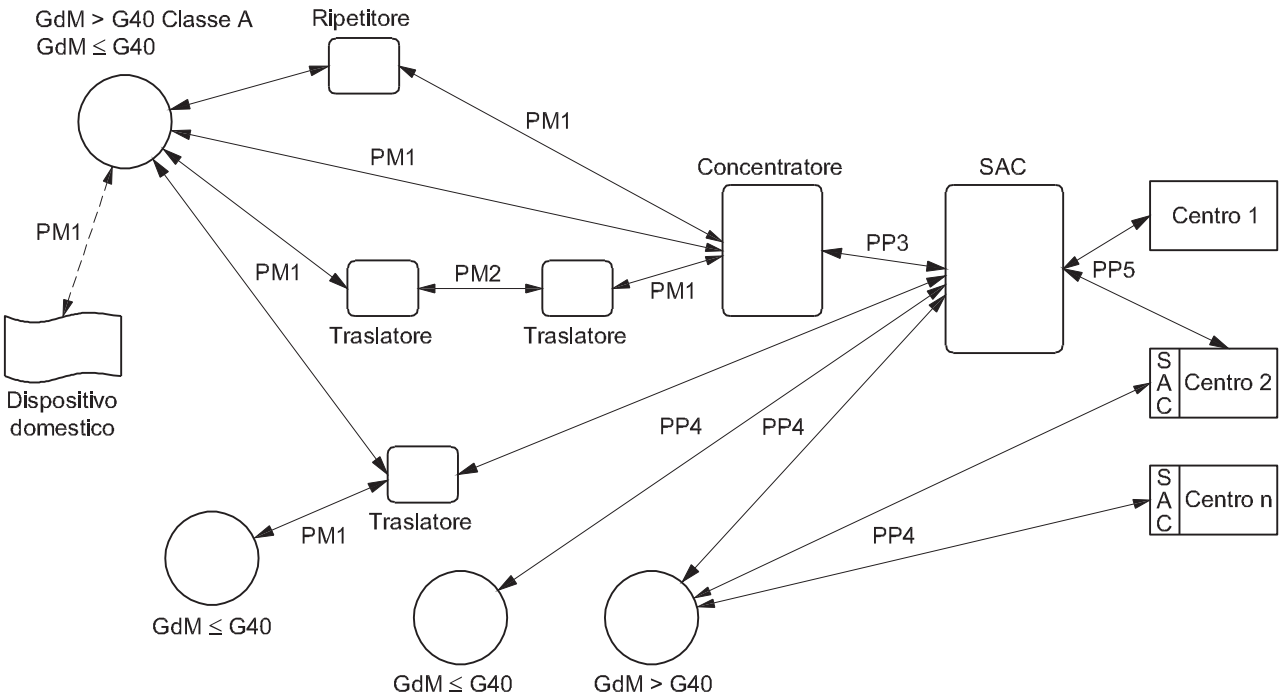
figure 1 **System architecture**



The system supports configuration with hierarchical and non-hierarchical centers.

In the hierarchical center configuration, only one center (SAC) is authorized to manage the network, i.e., to read and configure the GdMs and other system components; the other centers have access to the data and, where applicable, to the configuration only through the SAC.

In the configuration with non-hierarchical centers, all authorized Centers can access the GdMs according to rules that establish the authorizations for reading and configuration functions (Multi-access).

The GdMs can be managed directly by the SAC using specific protocols with point-to-point profile - PP4, reported in UNI/TS 11291-8 or they can be managed by the centers through concentrators.

The use of GdMs capable of communicating with the SAC in point-to-point or point-to-multipoint mode can occur without any constraints in relation to the technical/economic advantages attributable to the types of installation.

GdMs intended for measuring systems with GdM greater than G40 or range greater than 65 m^3/h at the reference thermodynamic conditions, can be read directly by the centers in the configuration of non-hierarchical centers and therefore, being able to manage multi-access, they must use point-to-point protocols with PP4 profile reported in UNI/TS 11291-8 and detailed in UNI/TS 11291-3 and UNI/TS 11291-4.

The Concentrator communicates with the GdMs it manages via wireless communication using the PM1 point-to-multipoint profile protocol defined in UNI/TS 11291-8. For this communication, the Concentrator can use Repeaters that do not make any modifications to the protocol levels, with the primary purpose of making it possible to overcome the distances separating the Concentrator from the GdM.

For communication between Concentrators and GdM, the architecture also provides for the possibility of using different protocols and transmission media as long as the following requirements are met:

- access to the PM1 profile wireless network must be done via translators;
- the protocol used must be one of those defined in UNI/TS 11291-8.

The SAC communicates with the Concentrators it manages, using the most suitable point-to-point transmission means and protocols with PP3 profile defined in UNI/TS 11291-8.

The SAC can communicate with the GdMs in point-to-point mode, using the most suitable point-to-point transmission media with PP4 profile or through translators.

4.2

Services

The remote reading and remote management system must be able to ensure support for the performance of the following functions (services):

- synchronization: the set of services that allow for a synchronized time reference base to be available in all system components, where necessary;

- fraud detection and reporting: the set of services that allow the detection, in all system components, of fraud and/or tampering attempts and the reporting of these to the relevant Centre;
- dynamic multi-tariff capacity: the set of services that allow gas consumption to be processed across structured tariff bands (tariff programs), making it possible to configure and reconfigure tariff programs, even remotely;
- supply management: the set of services that allow the interruption and enabling of the restoration of the supply in the GdM;
- consumer information: the set of services that allow communication with end-customer devices to provide information that is, among other things, useful for energy saving;
- software update: the set of services that enable the updating, even remotely, of the software of all the components of the System;
- measurement and load profiles: the set of services that allow the measurement of the quantities of gas supplied, their computation in organised registers (load profiles), and their transmission to the relevant Centre;
- infrastructure management and maintenance: the set of services that allow the detection of significant changes in the functioning of the System so that automatic or manual actions can be activated to restore optimal operating conditions;

- anomaly detection and reporting: the set of services intended for the detection of functional anomalies in the equipment and their reporting to the relevant Centers.
- security management: the set of services designed to configure the functions required to ensure the security and confidentiality of communications as well as access to information stored on network devices by unauthorized personnel.

4.3

Procedures for the performance of the Services

4.3.1 Service: Synchronization

The synchronization service serves two purposes: first, maintaining metrological and functional characteristics, and second, ensuring the functioning of the communication network. The minimum requirements for the second purpose are explained below, referring to the relevant standards governing metrological and functional aspects for those relating to the first.

The synchronization service must be performed by each component of the system on the components directly interfaced with it and dependent on it for synchronization in order to guarantee a residual error after synchronization.±2 s (see figure 2).

For each component of the system that implements clock functions, self-synchronization is permitted provided that it complies with the frequency and limits indicated in this point.

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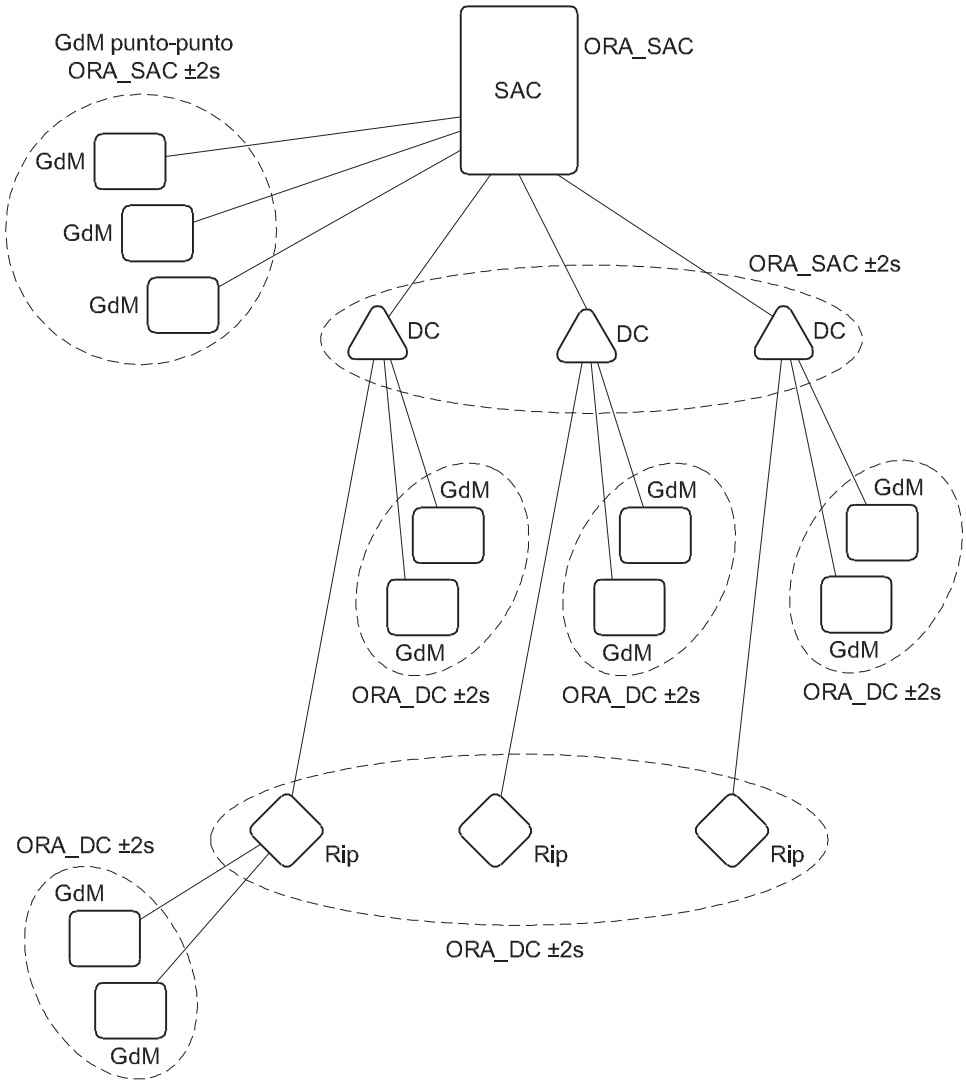
figure 2 **Synchronization**



Concentrator/Translator RTC Synchronization: The service, where applicable, must allow for the automatic synchronization of the Concentrator clock so that the misalignment between the Concentrator and SAC clocks is kept within 2 s; the service can be activated by the SAC at each connection to the Concentrator and in any case no later than 3 days after the previous synchronization; it must produce information in the SAC on the outcome of the synchronization within a time frame no greater than 24 hours from the moment of activation.

Repeater RTC synchronization: The repeater does not perform metrological activities; however, some services, such as software updates, require the management of a calendar clock, which requires a synchronization function. The synchronization service is provided by the Concentrator. The service must allow the Repeater's clock to be synchronized so that the misalignment between the Repeater's clocks and that of the Concentrator is kept within 2 seconds; the service can be activated by the Concentrator at each connection to the Repeater and in any case every 3 days at most; for this service, the Concentrator can use other Repeaters; it must provide the Concentrator with information on the outcome of the synchronization within a timeframe no greater than 3 days from the moment of activation.

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GdM RTC synchronization: For GdMs managed through a concentrator, the service must allow for the automatic synchronization of the GdM's clock so that the misalignment between the Concentrator and GdM clocks is kept within 2 s; the service can be activated by the Concentrator at each connection to the GdM and in any case every 3 days at most; for this service, the Concentrator may use Repeaters; it must provide the Concentrator with information on the outcome of the synchronization and on the GdM's diagnostic status (diagnostics, events, etc.) within a timeframe no greater than 3 days from the moment the service is activated.

For GdMs managed through the Translator, the service must allow for automatic synchronization of the GdM's clock so that the misalignment between the translator's and GdM's clocks is kept within 2 s; the service can be activated by the Translator at each connection to the GdM and in any case every 3 days at most; for this service, the Translator may use Repeaters; it must produce information on the synchronization outcome in the SAC within a timeframe no greater than 3 days from the moment the service is activated. If the Translator does not possess the GdM's encryption and authentication keys, synchronization can occur with messages authenticated and encrypted with the network key. In this case, synchronization of more than 60 s in a month is not permitted.

The service for GdMs managed directly by the SAC must allow for automatic synchronization of the GdM's clock so that, after synchronization, the misalignment between the Concentrator and GdM clocks is maintained within 2 seconds; the service is activated by the Concentrator at each connection to the GdM and in any case every 3 days at most. The service must provide the SAC with information on the outcome of the synchronization within a timeframe no greater than 3 days from the time the service is activated.

4.3.2 Service: Fraud Detection and Reporting

Fraud Detection: Every network device, including the GdM of any class and type, must be capable of detecting fraud attempts. To this end, the device must be capable of detecting at least the conditions listed in Table 1, where applicable.

prospectus 1 **Conditions relating to fraud attempts**

Code	Conditions	Precautions	Applicable to
1	Power failure	Devices must be able to verify that the power source has been removed	At least to the GdMs and concentrators, if the condition generates a stoppage of the measurement and/or calculation and/or recording functions
2	Disconnecting one or more physical modules	The connection between physical modules must be continuously verified or protected by seals	Devices made from multiple physical modules
3	Unauthorized attempt to configure or modify parameters	Every unauthorized attempt must be recorded in a special log.	At least to devices whose access is governed by a key
4	Attempt to log in with altered keys	Every attempt to communicate with an incorrect access key must be recorded in a special log.	At least to devices whose access is governed by a key
5	Application of external fields (magnetic, optical, etc.)	The fraudulent application of external fields to which the device is susceptible must be detected	GdM, if external fields can alter the measurement or calculation functions or registration
6	Access to electronic parts	Opening of guards to electronic parts must be detected or protected by seals	All devices on the network

Detection of the above conditions must result, where applicable, in the alarm being recorded in the event log. The recording must include at least:

- date and time of detection;

- standard condition code.

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4.3.3 Service: Dynamic multi-tariff capability

Tariff Schedule Configuration: The service must allow the configuration of a new Tariff Schedule in the GdM. The service is activated by the relevant SAC when necessary. The service must generate a status change in the GdM receiving the command (acquisition of the new pricing schedule) and inform the SAC that the command has been received within a period of no more than 7 days from the time the service is activated. The service only allows the configuration of the Future Tariff Schedule.

A new rate program configuration must be associated with a date indicating its effective date. This date must be defined at least as a day, as the effective date is expected to always coincide with the start/end time of the conventional gas day. The configuration of a new rate program must be tracked by the GdM in the event log; the effective date of a new rate program, in addition to being tracked in the event log, must also result in the closure of the current billing period.

Billing Period Configuration: The service must allow the configuration of a new Billing Period in the GdM. The service is activated by the relevant SAC when necessary. The service must generate a notification in the GdM indicating the activation of the new billing period and in the SAC indicating receipt of the command within a period no greater than 7 days from the time the service is activated. The configuration of a new billing period must be tracked by the GdM in the event log and must result in the closure of the current billing period.

Configuring the Commercial "Switch" date (change of sales company): the service must allow the configuration of a date in the GdM on which the end customer's commercial switch will be performed. The service is activated by the relevant SAC when necessary. The service must generate a status change (date acquisition) in the GdM receiving the command and inform the SAC of the receipt of the command within a period of no more than 7 days from the time of service activation. Reaching the scheduled switch date, in addition to being tracked by the GdM in the event log, must result in the closure of the current billing period.

4.3.4 Service: Supply Management

Supply management: The service, where applicable, must allow the valve to be closed and opened on the control units equipped with it. The service is activated by the relevant SAC when necessary. The service must generate a status change in the control unit receiving the opening enable command within 24 hours of receiving the user reconnection notification, and the SAC must inform the SAC that the command has been received within a period of no more than 24 hours from the time the service is activated.

The closing/opening of the valve must trigger an event that the GdM records in its event log.

The GdM must guarantee the actual closing or opening state of the valve:

- in the “closed” valve state the GdM must be able to implement a diagnostic activity aimed at verifying that there is no gas flow at least within the limits of the minimum measurable flow rate (see UNI/TS 11291-6);
- in the state immediately following the reopening of the valve, the GdM must be able to implement a diagnostic function that highlights any openings of the valve with a leak in the user's system (see UNI/TS 11291-6).

To ensure rapid reopening of the valve (for example, after a closure due to non-payment), the valve can be reopened by entering a secret code into the control panel (for example, by the user). If the secret code is detected as incorrect by the control panel a configurable number of times, the control panel must cancel the secret code, generate an event, and keep the valve closed while awaiting a subsequent reopening enable command sent by the SAC.

4.3.5 Service: Consumer Information

Data display on the GdM display: the GdM must be equipped with an electronic display; the service performed independently by the GdM must allow the display on the GdM display of at least the data indicated in Table 2.

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prospectus

2 Data display on the GdM display

Parameter	Notes
PDR	PDR Code
Date and time	Of the place where the GdM is installed that can follow daylight saving time
Diagnostics	The coding of which must be standardized for each GdM class
Valve status	For GdM classes that require it, the (normalized) reason for the "closed" status must also be indicated.
Totalizer of volumes at thermodynamic reference conditions	Current absolute[to)]
Current rate	
Totalizer of volumes at thermodynamic reference conditions	Current absolute[to)] in the F1 band
Totalizer of volumes at thermodynamic reference conditions	Current absolute[to)] in band F2
Totalizer of volumes at thermodynamic reference conditions	Current absolute[to)] in band F3
Closing date of the previous period	Equivalent to the start date of the current billing period
Totalizer of volumes at thermodynamic reference conditions	Absolute in band F1 at the end of the previous billing period
Totalizer of volumes at thermodynamic reference conditions	Absolute in band F2 at the end of the previous billing period
Totalizer of volumes at thermodynamic reference conditions	Absolute in band F3 at the end of the previous billing period
Message to the user	Message of up to 100 printable characters (optional and where required by the relevant parts of the UNI/TS 11291 series)

to) Updated at the last connection with the GdM.

The service is activated by the GdM when requested by the User and must produce the information on the viewer within a time no longer than 3 seconds from the moment of activation.

Data transmission for the User's remote device: the service, for the class of devices for which it is provided, must allow the sending to the User's remote device of at least the data shown in Table 3 below.

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prospectus 3 **Data transmission for remote device**

Parameter	Notes
Date and time	Of the place that can follow daylight saving time
Diagnostics	The coding of which must be standardized for each GdM class
Valve status	For GdM classes that require it, the (normalized) reason for the "closed" status must also be indicated.
Totalizer of volumes at thermodynamic reference conditions	Current absolute
Current rate	-
Totalizer of volumes at thermodynamic reference conditions	Absolute current in F1 band
Totalizer of volumes at thermodynamic reference conditions	Absolute current in band F2
Totalizer of volumes at thermodynamic reference conditions	Absolute current in F3 band
Last billing date	Of the closure of the billing period
Totalizer of volumes at thermodynamic reference conditions	Absolute in band F1 at the end of the previous billing period
Totalizer of volumes at thermodynamic reference conditions	Absolute in band F2 at the end of the previous billing period
Totalizer of volumes at thermodynamic reference conditions	Absolute in band F3 at the end of the previous billing period
Volumes	Consumed in the last 10 days
Message to the user	message of up to 100 characters

The service is activated by the GdM periodically for a configurable period (PRD) but in any case no longer than 12 hours.

Message display on the GdM display: The service must allow a message sent by the SAC to be displayed on the GdM display. The message must have the following characteristics:

- must be able to be displayed on a numeric display (7 segments);

- horizontal scrolling can be used if the message is longer than the display can accommodate;

- the message cannot be longer than 100 characters;

- a new message overwrites the previous one;

- each message has at least one display start date associated with it.

The service is activated by the relevant Center when necessary and must produce information in the SAC on the outcome of the GdM status change (message reception) within a period of no more than 72 hours from the moment of activation.

Message display on the remote viewer: The service must allow a message sent by the SAC to the GdM to be displayed on the User's remote viewer. The message, consisting of ASCII alphanumeric (printable) characters, cannot exceed 100 characters. Each message must have an associated validity period.

The service is activated by the relevant Center when necessary. The message is sent from the SAC to the GdM, then periodically from the GdM to the remote viewer with a PRD period and must provide the SAC with information on the outcome of the GdM's status change (message reception) within a period no greater than 72 hours from the moment of activation.

4.3.6 Service: Software Update

GdM Software Update: The service must allow for the GdM software to be updated. The service is activated by the SAC when necessary and must provide the SAC with information on the outcome of the GdM status change (acquisition of new software) within a period of no more than 45 days from activation.

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The GdM software update must be carried out in compliance with the provisions of Welmec Guide 7.2 and UNI/TS 11291-10.

The activation of new metrology software must be tracked in the event log.

Repeater/Translator Software Update: The service, where provided, must allow the Repeater or Translator software to be updated. The service is activated by the SAC when necessary and must provide the SAC with information on the outcome of the repeater/ translator status change (acquisition of the new software) within a period of no more than 45 days from activation.

Concentrator software update: The service, where provided, must allow for the Concentrator's software to be updated. The service is activated by the SAC when necessary and must provide the SAC with information on the outcome of the concentrator's status change (acquisition of the new software) within a period of no more than 5 days from activation.

4.3.7 Service: Measurement and load profiles

Reading of tariff registers: the service must allow the tariff registers processed by the GdM to be recorded in the SAC database; the service must be able to be activated by the SAC and/or the Concentrator and/or the GdM periodically with configurable intervals ranging from 1 to 180 days; it must produce information on the reading outcome in the SAC within a timeframe no greater than 7 days from the moment of activation.

Load curve reading: the service must allow the load curves developed by the GdM to be recorded in the SAC database; the service must be able to be activated by the SAC and/or the Concentrator and/or the GdM as needed and/or periodically with configurable intervals between 1 and 70 days; it must produce information on the reading outcome in the SAC within a timeframe no greater than 7 days from the moment of activation.

4.3.8 Service: Infrastructure Management and Maintenance

The services listed below refer to conditions in which GdM and Repeater type network devices can be installed, at the same time as or after the installation of the Concentrators/Translators.

Repeater Management: Repeaters are useful for extending the wireless communication range of GdMs that are unable to communicate directly with at least one Concentrator/ Translator in their network. With this in mind, the service, where applicable, must allow registration (affiliation).[1]of a new repeater in the database of at least one of the system's Concentrators/Translators; the affiliation service is activated during or after the Repeater's installation and must provide the SAC with information on the outcome of the acceptance within 3 days of the affiliation procedure being activated; for this service, the system may make use of other Repeaters already enrolled.

Management of the GdM by a Concentrator/Translator: the service, where applicable, must allow the registration of a new GdM in the database of at least one of the system's Concentrators (affiliation); the service can be activated by an authorized operator during the installation of the GdM (provided at least one Concentrator/Translator capable of managing the GdM is available), or it can be activated automatically and periodically by the GdM; it must produce information on the GdM and the Concentrator/Translator regarding the outcome of the affiliation within a period of no more than 3 days from the moment the service is activated.

The same GdM can be taken over (affiliated) by one or more Concentrators/Translators.

1) A device taken over by the device but not yet enrolled is considered affiliated.

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Enlistment[2]of a GdM: the service must allow the registration of a new GdM in the SAC database and the SAC to configure all the parameters necessary for the GdM's operation; the service is activated by the GdM for GdMs that communicate directly with the SAC, by the Concentrator/Translator for GdMs affiliated with it, or by the SAC if necessary. The service must provide the SAC with information on the outcome of the configuration within a timeframe no greater than 3 days from its activation. The GdM's configuration must generate an event that must be recorded by the GdM in its event log.

GdM reconfiguration: the service must allow remote modification of one or more GdM parameters, for which reconfiguration is provided; the service is activated by the SAC when necessary and must produce information in the SAC on the outcome of the GdM reconfiguration within a timeframe no greater than 7 days from the moment of activation.

Network reconfiguration: The service, where applicable, must allow for network reconfiguration in situations where the network structure is altered; the service can be activated by any of the system components (SAC and/or Concentrator and/or Translator and/or Repeater and/or GdM) when necessary and must provide the component that generated it with information on the outcome of the reconfiguration within a timeframe no greater than 7 days from the moment of activation.

Reading GdM status: the service must allow the SAC to acquire knowledge of the status of the individual GdM; the service must be able to be activated by the SAC when necessary and must cause the SAC to acquire the status of the individual GdM within a timeframe no greater than 7 days from the moment of activation.

The GdM status must include at least the following information:

- identifier of the tariff program active in the GdM;
- identifier(s) of the software active in the GdM;
- billing period in use in the GdM;
- number of recorded events;
- fraud attempt counter log;
- diagnostic data.

Repeater or Translator Status Reading: The service must allow the SAC to acquire knowledge of the status of each individual repeater or translator registered under a concentrator. The service must be able to be activated by the SAC when necessary and must cause the SAC to acquire the status of each individual repeater and translator within a timeframe no greater than 7 days from the moment of activation.

The status of the repeater or translator certified under a concentrator must include at least the following information:

- active software identifier(s);
- number of errors recorded;
- number of enrolled/affiliated devices
- diagnostic data.

Concentrator or Translator Status Reading: The service must allow the SAC to acquire knowledge of the status of the concentrator or translator that communicates directly with the SAC. The service must be able to be activated by the SAC when necessary and must cause the SAC to acquire the concentrator's status within a timeframe no greater than 3 days from the moment of activation.

The status of the Concentrator or Translator that communicates directly with the SAC must include at least the following information:

- identifier(s) of the software active in the concentrator;
- number of GdMs enrolled/affiliated;
- number of recruited/affiliated repeaters;

2) A device taken over by the SAC is considered to be enlisted.

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- number of recruited/affiliated translators;
- number of errors recorded;
- diagnostic data.

Concentrator/Translator Configuration Reading: The service must allow the SAC to acquire the configuration of the concentrator or translator that communicates directly with the SAC. The service must be able to be activated by the SAC when needed and must provide the SAC with information on the concentrator's status within a timeframe no greater than 3 days from the moment of activation.

The concentrator/translator configuration must include at least the following information:

- list of affiliated GdMs (the identifier is sufficient);
- list of affiliated repeaters (identification is sufficient);
- list of affiliated translators (identification is sufficient);
- communication reachability statistics for each component.

4.3.9 Service: Detection and reporting of anomalies

All system devices (including SAC) must include a diagnostic function that verifies the reliability of their database. If the device detects that a processed or recorded data item is unreliable and the correct value cannot be retrieved in any way, this condition must be reported so that anyone using the data can consider not using it.

GdM event management: the events that must be managed are the minimum ones specified in the relevant sections of the UNI/TS 11291 series; the service must make the events processed by the GdM available to the SAC.

The service can be activated by the SAC, and/or by the Concentrator and/or by the GdM; it must produce information in the SAC of new GdM events within 7 days of the service activation.

GdM alarm management: Alarms are events that, given their significance, must be generated by the GdM and acknowledged by the SAC within a guaranteed maximum time of less than 24 hours. The use of alarm management services is not part of the system's minimum requirements and is therefore not covered by this technical specification.

Concentrator Event Management Concentrator/Translator: The service must make the events processed by the Concentrator or the Translator connecting to the SAC in point-to-point mode available to the SAC. The events processed by the Concentrator are at least the minimum required by UNI/TS 11291-7.

The service can be activated by the SAC and/or the Concentrator at each connection; it must produce information in the SAC of new Concentrator events within 3 days of activation.

Concentrator/Translator alarm management: alarms are events that, given their significance, must be generated by the Concentrator or the Translator and acknowledged by the SAC within a maximum guaranteed time of less than 24 hours.

The use of alarm management services is not part of the minimum system requirements and is therefore not covered by this technical specification.

4.3.10 Service: Security Management

For services relating to safety management, please refer to UNI/TS 11291-10.

4.4 System Communication Strategies

Considering that some elements of the system (excluding the SACs) could be battery-powered and considering that communication between the system devices represents a significant element of energy consumption, it is defined that all the elements of the system can support "power saving" strategies based on the activation of the communication function only during configurable time windows, leaving the possibility of deactivating communication for the remaining period.

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The operation of the system with communications only during configurable time windows is based on the following assumptions:

- the devices (GdM, routers, translators and concentrators) can have different time windows as long as they are coordinated (for example the concentrator can have a larger time window than that of a single GdM);
- the communication protocol must ensure that the clocks of the various elements are aligned within the necessary precisions indicated above;
- the plan describing the communication time windows must define the communication frequency by establishing a daily plan that allows for the definition of the windows as reported in Table 4;
- transmissions from a sending device must take place within the time slots of the receiving device.

prospectus 4 **Daily plan**

Communication Frequency	
Always active	
Everything is fine "x" minutes and with duration "d"	$x > 5$; $d < x$
Activate in time slots according to a schedule valid for weekdays (Monday-Friday)	Maximum 4 per day; the pair must be defined for each band x, d
Active in time slots according to a schedule valid for holidays and days before holidays (Saturday, Sunday)	Maximum 4 per day; the pair must be defined for each band x, d
Note The daily plan is characterized by a period (expressed in minutes) and a duration (expressed in seconds).	

Data transmission between the SAC (or concentrator) and the GdM is balanced, meaning communication can be initiated either by the SAC (or concentrator) or by the GdM.

If transmission begins from the GdM, the frequency and type of data to be transmitted must be indicated. In this case, the transmission frequency must at least include the types indicated in Table 5.

prospectus 5 **Transmission frequency**

Monthly Plan	
Transmission	Notes
Active every day	
Active only one day of the month	
Activate every x days	$x = 2 - 7$
Activate on the 1st and 15th of the month	
Daily Plan	
Activate after x minutes from the end of the conventional gas day	$x \geq 1$
Active only in communication bands 1/2/3/4	

4.5 Types and Classes of Devices

The elements that make up the remote reading or remote management system are each characterized by "logical" (not necessarily "physical") ports used for communication with the other system elements. Each logical port is characterized by a protocol profile that can include one or more protocols defined at all levels of the stack. If multiple protocols are provided for in a protocol profile, at least one of them can be implemented for that logical port.

There are two different types of protocol profiles:

- point-to-point (identified with the abbreviation PPn) which allow communication between individual devices;
- point-to-multipoint (identified with the abbreviation PMn) which allow communication between a single device and multiple devices.

The protocol profiles that can be used are described in UNI/TS 11291-8.

The system devices, in relation to the function they perform, are:

Centers: responsible for configuring and reconfiguring the GdMs and to which the consumption and diagnostic data processed by the GdMs are transmitted. For communication with the GdMs, they use SACs, which can be centralized or integrated into the Centers.

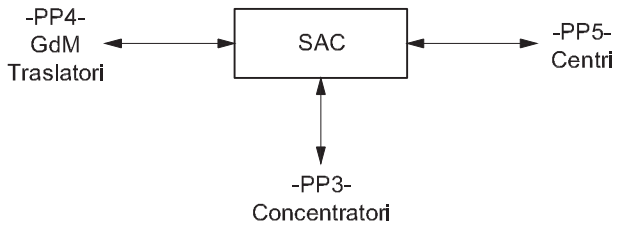
Central Acquisition Systems (CAS): These create communication interfaces between the Centers and the GdMs (see Figure 3). The CASs independently perform network and GdM management functions (e.g., synchronization, periodic data acquisition) and activate functions upon specific request from the Centers (e.g., reconfiguration of a GdM).

They implement PP4 profile point-to-point protocols towards the GdMs. They

implement PP3 profile point-to-point protocols towards the Concentrators. They

implement PP5 profile point-to-point protocols towards the Centers.

figure 3 **Central acquisition systems**



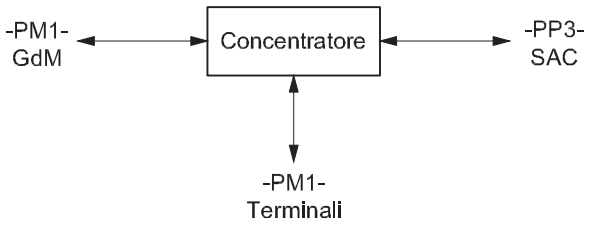
Concentrators: they concentrate the data from multiple GdMs, making them available to the SAC; they are the communication interface of the SAC towards the GdM (see figure 4).

They implement PP3 profile point-to-point protocols towards SACs. They

implement PM1 profile point-to-multipoint protocols towards GdMs.

They implement PM1 profile point-to-multipoint protocols towards the configuration and maintenance terminals.

figure 4 **Concentrator**



Repeaters and Translators: These are auxiliary network elements that facilitate communication between GdMs and Concentrators; they process the information required for communication but do not store data processed by the GdMs. They do not manage the GdMs' operating encryption keys (see Figure 5).

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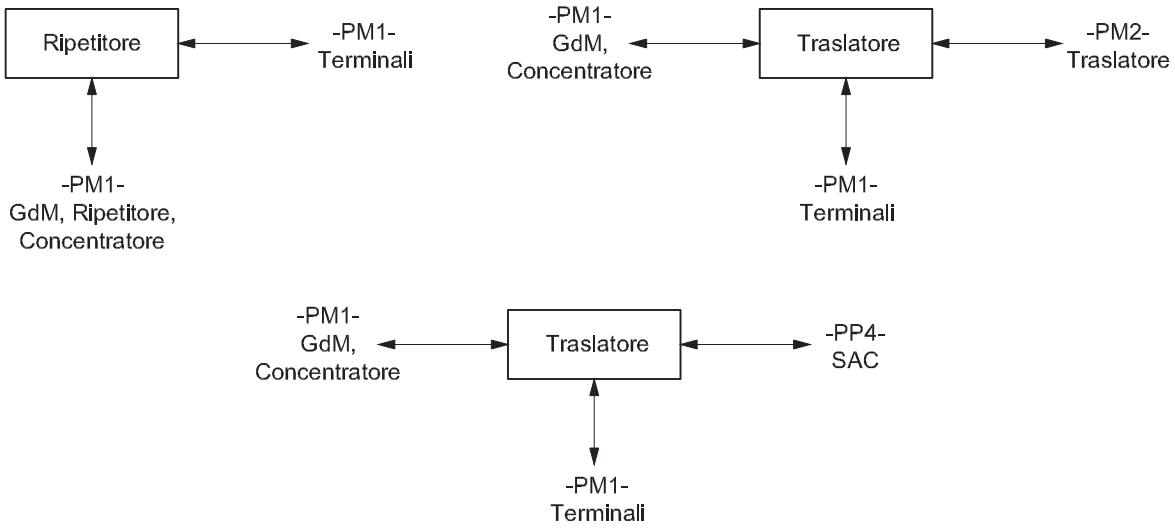
figure 5 **Repeaters and Translators**



Measurement Groups (GdM): These are divided into classes A1, A2, A, B, and C based on their functional characteristics and the data they can process. Depending on their class, they can communicate with concentrators (including through repeaters and translators) using point-to-multipoint protocols, or they can communicate directly with SACs using point-to-point protocols (see Figure 6).

figure 6 **Measuring Groups**

Communication between all elements of the system, regardless of the transmission medium and protocol used, must ensure high levels of protection for the transmitted data.

Therefore, all communications between system elements, unless otherwise specified, must take place in authenticated mode according to the AES 128 standard and the communication protocols must adopt specific techniques for encryption and recognition of data alterations during transmission.

4.6 Measurement Groups (GdM)

The GdMs operating in the remote reading or remote management system covered by this technical specification are divided into classes based on their field of application (meter class or maximum measurable flow rate), the functions they are required to perform, and the data they process.

GdMs with a meter class strictly greater than G40 or capable of measuring flow rates at reference thermodynamic conditions greater than 65 m3/h are divided into three classes indicated in the context of this specification with A, B, and C in relation to the functions and data processed. The functional and implementation specifications of the aforementioned GdMs are reported in UNI/TS 11291-4.

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GdMs with a meter class lower than or equal to G40 and higher than or equal to G10 or capable of measuring flow rates at reference thermodynamic conditions between 16 m^3/he 65 m^3/h included, are indicated in the context of this specification with class A1. The functional and construction specifications of the aforementioned GdMs are reported in UNI/TS 11291-5.

GdMs with a meter class lower than G10 or capable of measuring flow rates at reference thermodynamic conditions lower than 16 m^3/h are indicated in the context of this specification with class A2.

GdMs of all classes support time-window communication strategies.

For the purposes of the functionality of the GdMs in the remote reading or remote management system, the following is expected.

The A2 class GdMs:

- support remote management (remote control for valve management);
- must support a PM1 profile protocol or a PP4 profile protocol;
- they support communication with terminals being able to use the same physical port and the same PM1 or PP4 profile protocol;
- they do not support multi-access;
- must provide for communication to a User device;
- can communicate with the SAC through the Concentrators, either directly or through translators.

The A1 class GdMs:

- they only support remote reading;
- must support a PM1 profile protocol or a PP4 profile protocol;
- they support communication with terminals being able to use the same physical port and the same PM1 or PP4 profile protocol;
- they do not support multi-access;
- can communicate with the SAC through the Concentrators either directly or through translators.

Class A GdMs:

- they only support remote reading;
- must support a PM1 profile protocol or one of the PP4 profile protocols;
- they support communication with terminals being able to use the same physical port and the same PM1 or PP4 profile protocol;
- can communicate with the SAC through the Concentrators, or directly.

Class B and C GdMs:

- they only support remote reading;
- support one of the PP4 profile protocols;
- support communication with terminals using a dedicated physical port and one of the PP4 profile protocols;
- support multi-access;
- communicate directly with the SAC.

Table 6 summarises the relationship between the Classes of Measurement Groups, the class of the meter associated with the GdM, and the main characteristics of the communication.

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prospectus 6 **Class of GdM**

Class of GdM	A2	A1	TO	B	C
Main features					
GdM with counter < G10	X				
GdM with G10 counter≤GdM≤G40		X			
GdM with counter > G40 (hourly profile)			X	X	
GdM with counter > G40 (15 min profile)					X
Remote management	X				
Communication to user device	X				
Multiple access				X	X
PM1 protocol (point-multipoint)	X	X	X		
PP4 (point-to-point) protocol	X	X	X	X	X

If the GdM, of any class, is made with (physical) devices that each implement one or more functional modules, it must be possible to detect the integrity of the connection between the devices, generating an alarm if it is not verified. Furthermore, if the connection between the devices involves data communication, the manufacturer and/or integrator must provide evidence of the protection and control mechanisms adopted (seals, encryption, etc.) to ensure the non-alterability of the sampling data.

4.7

Repeaters and Translators

The repeat function is used in the remote reading or remote management system to extend the physical length of the network without modifying the transmission protocol.

The repetition function, supported by the protocol, can be implemented on specific network devices (Repeaters) or be implemented in the i-th GdM.

Repeaters and Translators support time-window communication strategies.

Repeaters used to extend the physical length of the network between two Translators are not covered by this technical specification.

Like Repeaters, Translators can also be used to extend the physical length of the network but, unlike Repeaters, Translators perform a transformation of the communication protocol.

The use of Translators in the remote reading or remote management network can be justified by:

- use of more reliable and/or economical and/or available transmission means;
- openness towards technological innovations that will inevitably occur during the life of the system;
- use of communication technologies that allow the physical length of the network to be extended.

Both Repeaters and Translators may need to be configured to function. In this case, configuration can be done using dedicated terminals that, for interoperability reasons, use the PM1 profile physical port and protocol. Autoconfiguration of the devices is permitted, meaning Repeaters and Translators can adopt strategies to define their correspondents in the communication network.

4.8

Concentrator

The Concentrator device in the remote reading or remote management network has the purpose of:

- manage communications with the SAC on one side and with the GdM on the other, through any repeaters/translators;
- make all management of the network on which the GdMs are installed and underlying it transparent to the SAC, if requested, so that the SAC can manage peripheral devices by addressing them only at the application level;

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- provide the SAC with complete visibility of the operating configuration of the network controlled by the Concentrator, informing it of any updates that have occurred;
- ensure system interoperability with peripheral devices and SACs supplied by different manufacturers.

The concentrator does not manage multi-access and therefore recognizes only one SAC in the network.

4.9 SAC

The SAC function in the remote reading or remote management system is the part of the center responsible for managing the relevant network.

All centers that have the right to communicate with the GdMs in any capacity must do so exclusively through the SAC, which is the only component of the system delegated to this function.

The SAC function can be physically separate from the rest of a center's functions (data storage, analysis, editing, etc.) or it can be integrated.

A remote reading or remote management network can have multiple SACs.

The SAC function in the remote reading or remote management system is responsible for managing the network in terms of:

- configuration/reconfiguration of all network devices (GdM, Repeaters, Translators);
- automatic and periodic acquisition of data processed by the GdMs via direct connection or by the Concentrators;
- acquisition of events and alarms processed by all devices on the network;
- Clock synchronization of directly managed GdMs and Concentrators. The SAC for managing concentrators is unique for each network.

The SAC must implement mechanisms for authenticating centers that have access rights.

The SAC must implement mechanisms for authentication and identification of Concentrators.

Data exchanged between SACs and Concentrators, as well as data exchanged between SACs and Centers (when these functional modules are not integrated), unless otherwise specified, must be encrypted.

4.10 Field start-up and maintenance operations

For the installation of equipment and any electrical and pneumatic connections, the manufacturer's instructions and the criteria set forth in applicable standards must be followed. Ease of inspection and calibration must also be ensured where required. When, for specific reasons, equipment must be installed outdoors, unless otherwise specified by the manufacturer, it must be housed within suitable protective enclosures such as cabinets, containers, or boxes. These enclosures must be constructed of suitable materials, of appropriate dimensions, and with measures to prevent temperature variations outside the limits established by the manufacturer.

During the initial installation and start-up of the remote reading or remote management system components, if applicable, the programming and initialization parameters required for their operation must be entered. The interfaces used for communication with the tools used for entering programming parameters and for maintenance are standardized for all components (including at the physical level).

5 TECHNICAL DOCUMENTATION

For all components of the remote reading or remote management system, the manufacturer must provide a suitable use and maintenance manual, which must declare the value of the optional parameters that can be selected and knowledge of which is deemed essential for the operation of the remote reading or remote management system.

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6 REQUIREMENTS RELATING TO LEGAL METROLOGY

Each device in the remote reading or remote management system must comply with current legislation and, in particular, with legal metrology requirements, where applicable. If one or more functional modules are implemented in the same device, they must comply with current legislation and, in particular, with legal metrology requirements, where applicable. In the event of overlap between the requirements of different legislative provisions, the most stringent requirement must be adopted.

In the absence of specific legislative provisions, the devices that perform the data recording function must comply with the applicable requirements of UNI EN 12405-1:2007.

As regards the functions relating to data recording and communication, in the absence of specific legislative provisions, the manufacturer can certify their performance, provided that:

- operates in compliance with UNI EN ISO 9001;
- provide evidence of the tests carried out to verify the requirements described in the relevant parts of the UNI/TS 11291 series.

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BIBLIOGRAPHY

Resolution ARG/gas 155/08 Guidelines for the commissioning of gas metering units, characterised by minimum functional requirements and with remote reading or remote management functions, for delivery points of natural gas distribution networks

UNI CEI EN 13757-2	Communication systems for meters and remote meter reading - Part 2: Physical and link layers
UNI CEI EN 13757-3	Meter communication and remote meter reading systems - Part 3: Dedicated application layer
UNI CEI EN 13757-4	Meter communication and remote meter reading systems - Part 4: Wireless meter reading (radio meter reading for operation in the SRD band 868 MHz to 870 MHz)
UNI EN ISO 5167-1	Measurement of fluid flow rate using differential pressure devices inserted into solid circular cross-section ducts - Part 1: General principles and requirements
UNI EN ISO 12213	Natural gas - Calculation of the compression factor
CEI EN 60870	Remote control systems and equipment
IEC EN 62056-21	Electricity metering - Data exchange for meter reading, tariff and load control - Part 21: Direct local data exchange
IEC EN 62056-46	Electricity metering - Data exchange for meter reading, load tariff control - Part 46: Data link layer using the HDLC protocol
IEC EN 62056-53	Electricity metering - Data exchange for meter reading, load tariff control - Part 53: COSEM application layer
IEC EN 62056-61	Electricity metering - Data exchange for meter reading, load tariff control - Part 61: Object Identification System (OBIS)
IEC EN 62056-62	Electricity metering - Data exchange for meter reading, load tariff control - Part 62: Interface classes
EIA/TIA-232-E	Interface Between Data Terminal Equipment and Data Circuit-Terminating Equipment Employing Serial Binary Data Interchange - Jan. 1991 (EIA: Electronic Industry Alliance; TIA: Telecommunications Industry Association)

NTA 8130 - Dutch standard for smart metering

Remote reading protocol - Snam Rete Gas/Misura

High Speed Data Collection (Metering International - 4/2003)

Customers Win With Internet Metering - Cost Savings and Pro.ts (Metering Int.- 4/2003)

Internet-based Least Cost Metering (Metering International - 3/2003)

Standardization and security in message exchange (Metering International - 3/2003)

Welmec 7.2 Software Guide (Measuring Instruments Directive 2004/22/EC)

(NIST FIPS PUB 197) Advanced Encryption Standard (AES)

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